assignment_03

Bill Perry

Ecological Statistics Assignment 03

The goal of this assignment is to test whether crayfish chelae (claw) length differs between upstream and downstream locations in six different rivers where waterfalls separate them. We hypothesize that downstream crayfish will have larger chelae due to increased food availability and different selective pressures.

Yo goal is to use an ANOVA of some sort to analyze this data.

There are 16 columns in the dataframe:

1. crayfish_id - crayfish ID sampled
2. river - the river it was collected form
3. position - upstream downstream
4. chelae_length - length of chelae

Your primary task is to answer the following questions:

Your task is to use the dataframe to examine patterns in crayfish chelae lenght in the different rivers upstream and downstream using some form of an ANOVA.

To complete the task you will submit two files

1. one containing your write-up
2. another containing fully functional and annotated R code that was used to generate summary statistics, figures and perform analyses.

Your write-up will include the following sections:

Your write-up will include the following sections:

Hypothesis statements

- In this section you will state the hypotheses you are testing in this assignment in verbal form as well as in mathematical symbol notation.

Statistical methods description

- This section will include a detailed description of the statistical analyses you performed:
  - assumptions of tests
  - how these assumptions were checked
  - name and version of the software used to perform analyses (R and the libraries used)

Statistical results description

In this section you should present a description of the data, report the main trends and results of statistical analyses (together with information about results of assumption testing) in text form (use standard format for reporting results of tests) and refer the reader to any figures and tables summarizing the data. Each figure or table presented **must** be explicitly mentioned in this section, with a short description of the information contained in the table or figure. Remember to include statements on what your results mean for the hypotheses you are testing.

Tables and figures

Here you will place all the figures and tables mentioned in the text. Tables go first, followed by figures. Tables and figures should be placed in the same order in which they are mentioned in the body of the text. Each table or figure should be accompanied by an informative caption that will allow the reader to understand what is shown in the table/figure without reference to the text. Captions go below figures and above tables.

Appendix

This section will contain any additional figures (e.g., residual and qq plots). All these figures should also be referred to in the text in the appropriate order.

Assignment Tasks:

- State the question and hypotheses using words and math
- State the type of test you are doing and why
- State the assumptions of the test
- Show the model statement you chose and what it means
- show the assumption tests in the appendix
- Do the post f tests of the model
- present the methods and results
- with final figures for publication

Grading Rubric:

Total is 100 points

- Hypotheses Statements - 10 points
- Statistical Methods - 20 points
- Results Statement - 30 points
- Figure and Table Quality - 20 points
- R Code - 16 points
- Grammar/Writing - 4 points